CIS 635: Assignment 2b

Supervised Learning II: Classification and Feature Importance

Deadline: Please check the Blackboard post **Total points:** 5

Objective

This assignment provides hands-on practice with **classification**, the supervised task of predicting a categorical label. You will build and compare three models: **logistic regression**, **decision tree**, and **random forest**. You will also **optimize the hyperparameters** of one of them with cross-validation and **identify the most important features**. The data are highly imbalanced, so choosing appropriate evaluation metrics is a central part of the assignment. The data are samples; do not make global conclusions from these samples.

Data

Use the following file:

- **Credit card transactions:** `https://storage.googleapis.com/download.tensorflow.org/data/creditcard.csv`
  Target: `Class` (1 = fraudulent, 0 = legitimate). Features: `Time`, `Amount`, and anonymized principal components `V1`–`V28`.

General rules. Hold out a *stratified* test set (e.g., 80/20) *before* any tuning or feature selection, and use it only for the final evaluation. Fix a random seed so your results are reproducible. Any scaling or feature selection that learns from the data must be fit on the training data only (use a `Pipeline`). Report **precision, recall, and F1-score for the fraud class**, along with a **confusion matrix**, for every model.

Runtime tip: The dataset has about 285,000 rows. If hyperparameter search is too slow, you may tune on a *stratified* random sample of the training data (keep the fraud ratio unchanged; state the sample size), use `n_jobs=-1`, and keep the grid small. Final models must still be evaluated on the full held-out test set.

Tasks

(a) Data Preparation — 0.5 points
Report the number and percentage of fraudulent transactions. Briefly explain what this class balance implies for model evaluation, and why *accuracy* is not a suitable metric here.

(b) Building the Three Models — 1.5 points
- (i) Fit a **logistic regression** model with feature scaling. (0.5)
- (ii) Fit a **decision tree** classifier. (0.5)
- (iii) Fit a **random forest** classifier. (0.5)

Default or simple hand-picked settings are fine here; report the test-set metrics for each.

(c) Hyperparameter Optimization — 1 point
Choose **one** of the three models (the random forest is recommended) and optimize **at least two** of its hyperparameters with at least three values each (e.g., `n_estimators` and `max_depth` for a random forest; `max_depth` and `min_samples_leaf` for a decision tree; `C` and `class_weight` for logistic regression). Use **5-fold stratified cross-validation** on the training data with F1 as the scoring metric. Report the best hyperparameters, the **mean and standard deviation** of the cross-validated F1-scores for the best setting, and the test-set metrics of the tuned model.

(d) Model Comparison — 0.75 points

Summarize all models (including the tuned one) in one table. Which model performs best, and is the improvement from tuning meaningful given the cross-validation standard deviation? Comment on the trade-off between false positives and false negatives in the fraud setting.

(e) Feature Importance — 1.25 points

(i) Identify the most important features using **two** different approaches:

- the built-in (impurity-based) feature importance of your random forest or decision tree, and
- a second approach: *permutation importance* computed on the test set, or the absolute coefficients of the scaled logistic regression model.

Plot the top 10 features for each approach and comment on how much they agree. Why might the two approaches disagree? (0.75)

(ii) Retrain your best model using only the top 5 and top 10 features (ranked on the training data). Compare the F1-score with the full feature set. What does this tell you about how many features are really needed? (0.5)

Submission Guidelines

Submit a runnable notebook (Python, R, or another programming language of your choice) through Blackboard (the link of your notebook, which stays in your Google drive or else where, is not accepted). Include sufficient markdown/comments to explain your analysis and decisions. Use appropriate labels, titles, units, and legends in visualizations. Report numerical results to three decimal places where appropriate, and report the random seed(s) you used.

Important: The goal is not to find the single highest score. Focus on a sound modeling process: use **proper train/test separation, cross-validation, appropriate metrics, and clear interpretation** to support your conclusions.